CURRICULUM VITAE

Niki Zhong Petrakos
niki.petrakos@mail.mcgill.ca

EDUCATION AND TRAINING

DEGREES EARNED

PhD – Biostatistics — Oct. 2026
McGill University; *Montreal, QC*
Thesis: Synthetic data generation for optimal randomized controlled trial design
Supervisor: Dr. Erica E. M. Moodie

Master of Science – Biostatistics — Mar. 2022
University of Washington; *Seattle, WA*

Bachelor of Science – Mathematics — Jun. 2020
McGill University; *Montreal, QC*

RESEARCH EXPERIENCE

Postdoctoral Fellow — Oct. 2026 - Present
Chronic Viral Illness Center, Research Institute of the McGill University Health Centre, *Montréal, QC*
Research Advisor: Dr. Alexandra de Pokomandy, Professor & Physician

- Design and lead a research study investigating the real-world effectiveness of statin primary prevention of cardiovascular disease among people living with HIV in the province of Quebec
- Deliverables: knowledge translation (conference presentations, peer-reviewed publications), community engagement
- Consult as biostatistician on various research projects involving the Quebec HIV Cohort (the largest data cohort of people living with HIV in the province)
- Funded by the Canadian Trials Network (CTN+)

Research Assistant — Sep. 2022 – Sep. 2026
Chronic Viral Illness Center, Research Institute of the McGill University Health Centre, *Montréal, QC*
Research Advisor: Dr. Alexandra de Pokomandy, Professor & Physician

- Collaborated with family medicine clinicians to develop and execute principled, robust, and interpretable statistical analyses for a range of studies related to HIV, Hepatitis C, and primary care
- Produced clear, well-commented R code that is shared with the greater research team and that enables future seamless data synthesis (e.g., developed algorithm to create chronological multi-drug ART regimens using messy HIV EHR records)
- Worked independently and collaboratively with researchers of different fields, presenedt results to pharmaceutical partners

Graduate Researcher — Sep. 2021 – Mar. 2022
Public Health – Seattle & King County, *Seattle, WA*
Research Advisor: Dr. Alastair Matheson, Epidemiologist

- Collaboratively with three biostatistics master's students, planned and conducted statistical analyses in R to study the relationship between positive and negative exits from public housing and time to experiencing homelessness within one year post-exit, thus providing quantitative results to help inform public housing policy (work resulted in a published peer-reviewed article)

Undergraduate Researcher Jan. 2019 – Feb. 2020
McGill University, *Montréal, QC*
Research Advisor: Dr. Russell Steele, Professor

- To support replicable research, ran simulation studies to replicate results from published papers that studied Type I error rate control methods (multiple hypothesis testing, cumulative meta-analysis)

Research Assistant Summer 2017, Summer 2018
University of Washington Medical Center, Harborview Medical Center, *Seattle, WA*
Supervisor: Dr. Charles Landis, Principal Investigator & Physician

- Informed consent, RedCap data management for a range of liver disease studies; linked patients to Hepatitis C care by enrolling hard-to-reach patients (street-based, history of injection drug use)

PROFESSIONAL EXPERIENCE

Co-Instructor: Epidemiology and Data Analysis in Primary Care Sep. – Dec. 2026
McGill University, *Montréal, QC*
Supervisor: Dr. Alexandra de Pokomandy, Professor (FMED 505)

- Give 3-hour lectures on fundamental statistical concepts and R coding to equip students from a range of medical education backgrounds with the necessary skills to conduct biomedical and epidemiological research; design homework assignments

Teaching Assistant: Epidemiology Regression Methods Jan. – Apr. 2026
McGill University, *Montréal, QC*
Supervisor: Dr. Robert Platt, Professor (BIOS 602)

- Lead discussion sections about journal articles and state-of-the-art quantitative methods, create answer keys and grade assignments

Teaching Assistant: Quantitative Research Methods Aug. – Dec. 2024
McGill University, *Montréal, QC*
Supervisor: Dr. Nandini Dendukuri, Professor (EXMD 634)

- Supported graduate student learning by creating assignment solutions, holding weekly office hours, and explaining statistical concepts in an accessible manner

Marketing Analytics – Graduate Intern Jun. 2021 – Aug. 2022
Kaiser Permanente, *Oakland, CA*
Supervisor: Smriti Malla, Data Scientist IV

- Delivered interpretable results to business partners by quantifying care utilization impact (telehealth v. in-person) on patient retention; measured impact of B2B campaigns through weekly dashboard maintenance; worked collaboratively in a team of statisticians

Teaching Assistant Jul. 2021
UW Summer Institute in Statistical Genetics, *Seattle, WA*
Supervisor: Dr. Ken Rice, Professor

- Supported University of Washington (UW) Biostatistics instructors in the Intro to R short course by leading breakout room sessions and guiding students through practice problems

Math Instructor Jul. 2020 – Dec. 2020
Mathnasium, *Sammamish, WA*
Supervisor: Sumudini Guyer, Center Director

- Worked in groups and one-on-one with K-12 students to improve their math skills, gain confidence, and learn to love math

LEADERSHIP AND COMMUNITY SERVICE

Co-Founder Mar. 2025 – present
The Action Pact, *Montréal, QC*
- Created with two other women The Action Pact: a digital platform and civic engagement initiative to empower Canadians with tools, knowledge, and connections for informed, strategic civic action
- Developed a predictive model for the backend of a strategic voting tool, utilizing historical election data from the province of Ontario
- Led, in English and French, strategic voting experiential workshops for high school and undergraduate students at the Loran Scholarship Nationals Selection event
- Co-hosted seasonal book club events in Montréal, QC

Volunteer Graduate Student Mentor Sep. 2023 – present
Epidemiology, Biostatistics, and Occupational Health Buddy Program, McGill Scientista Mentorship Program, *Montréal, QC*
- Mentored first-year graduate and upper-year undergraduate students to help them integrate in the department and navigate graduate school and job applications

Co-Organizer May 2026
McGill University (Bio)Statistics Research and Career Day, *Montréal, QC*
- With 3 other PhD students, organized the 9th annual edition of a research and career day hosted by the biostatistics and statistics departments at McGill University
- Primary organizer for 3 keynote speakers (statistics professors across U.S. and Canada), 4 career panelists (McGill biostatistics and statistics alumni), venue and catering
- Collaborated with graduate students as well as faculty, departmental staff, venue staff, and funding agencies (secured $)

Biostatistics Program Student Representative Sep. 2023 – Sep. 2024
Epidemiology, Biostatistics, and Occupational Health Student Society, *Montréal, QC*
- As an elected council member, represented biostatistics student interests at departmental faculty meetings and informed students of decisions; organized events for biostatistics students, faculty, and staff to foster greater departmental camaraderie

Biostatistics EDI Committee Member Sep. 2020 – Sep. 2021
Biostatistics Equity, Diversity, Inclusion (EDI) Committee, *Seattle, WA*
- As a member of the Outreach and Pipeline Building Subcommittee, collaborated with community organizations to support disadvantaged high school students in their college endeavors

President May 2019 – Apr. 2020
Vice President – Fundraising Sep. 2018 – Apr. 2019
Vice President – Social Media Sep. 2017 – Apr. 2018
Alegria Contemporary Ballet Company, *Montréal, QC*
- Produced shows, organized weekly rehearsals, cultivated a welcoming dance environment; made ballet more accessible to the Montreal community via affordable classes and discounted tickets
- Organized fundraisers to ensure Alegria's financial security and ability to produce performances
- Advertised events and fundraisers, designed social media posts, cover art, and posters, and created promotional videos

PUBLICATIONS

Petrakos, N. Z., Moodie, E. E. M., Savy, N., de Raad, M., Belzile, E., and Lambert, S. (2026). Simulation-based power analysis for sequential multiple assignment randomized trials. https://arxiv.org/abs/2607.21751

Petrakos, N. Z., Moodie, E. E. M., and Savy, N. (2025). Incorporating missingness in a framework for generating realistic synthetic randomized controlled trial data. https://arxiv.org/abs/2512.00183.

Petrakos, N. Z., Moodie, E. E. M., and Savy, N. (2025). A framework for generating realistic synthetic tabular data in a randomized controlled trial setting. *Statistics in Medicine*, 44(18-19): e70227, https://doi.org/10.1002/sim.70227.

Petrakos, N. Z., Liu, Z., Hu, H., Keating, T., Colombara, D. V., Laurent, A. A., Chan, A., Pennucci, A., Matheson, A. I. (2025). Associations between exit type from federal housing assistance and subsequent homelessness. *Housing Policy Debate*, *35*(6), 966-984. https://doi.org/10.1080/10511482.2023.2243260.

Dolmatov, M.*, **Petrakos, N. Z.***, Moodie, E. E. M., Thomas, R., Durand, M., Klein, M., De Pokomandy, A. (2024). "Regression-based estimation of optimal adaptive treatment strategies: Key methods." In *Biostatistics in Biopharmaceutical Research and Development: Clinical Trial Analysis, Volume 2*. https://doi.org/10.1007/978-3-031-65937-9_12.

*Co-first authors

PRESENTATIONS

Oral Presentations

Epidemiology, Biostatistics and Occupational Health Student Seminar, *Montréal, QC* — Sep. 2026

- Gave an hour-long presentation about two first-author papers from doctoral research work to a graduate student audience

56es Journées de Statistique de la Société Française de Statistique, *Marseille, France* — Jun. 2025

- Presented first-author paper about frameworks to generate complex synthetic trial data to an audience of statisticians

(Bio)Statistics Research Day, *Montréal, QC* — May 2024

- Presented original research in synthetic data generation for clinical trials to an audience of biostatistics/statistics students and faculty

(Bio)Statistics Research Day, *Montréal, QC* — May 2023

- Presented culminating work from master's degree to an audience of biostatistics and statistics students and faculty

Capstone Presentation, *Seattle, WA* — Mar. 2022

- Presented master's capstone project alongside group members to an audience of biostatistics and statistics students and faculty

Poster Presentations

Scientific Days Réseau Santé Numérique – Poster Competition, *Quebec City, QC* — Mar. 2026

- Presented doctoral research about methods to account for missing data in the generation of realistic and complex synthetic trial data

Statistical Society of Canada Annual Meeting – Poster Competition, *Ottawa, ON* — May 2023

- Presented scientific questions, methods, and results for case study competition, optimizing patient flow in emergency departments

HONORS AND AWARDS

First Prize PhD Oral Presentation ($125) – Canadian Statistics Student Conference	May 2026
Second Place Poster Competition ($150) – Réseau Santé Numérique	Mar. 2026
Graduate Mobility Award ($7,426) – McGill University	Jan. – Jun. 2025
Mitacs Globalink Research Award for Research Abroad ($6,000) – Mitacs	Jan. – Jun. 2025
Research Internship Funding Award ($9,000) – Centre de Recherches Mathématiques	May 2024
Student Presentation Award, Second Place ($75) – 7th Annual (Bio)Statistics Research Day	Feb. – Apr. 2024
Graduate Mobility Award ($4,400) – McGill University	Feb. – Apr. 2024
Research Internship Funding Award ($7,500) – Centre de Recherches Mathématiques	
Honorable Mention Poster Competition ($125) – Statistical Society of Canada Annual Meeting	May 2023
School of Public Health Excellence Award - Outstanding Master's Student, Biostatistics – University of Washington	May 2022
James McGill Scholarship ($5,000) – McGill University	Sep. 2016

GRANTS

CIHR Project Grant Competition – Fall 2025 ($100,000 bridge funding) Mapping and predicting patient trajectories for acute respiratory infections and related antibiotic use with the multisite COhort in PRImary Care (COPRI) Principal Investigators: de Pokomandy A (nominated), Gaboury I, Paquette JS, Rodrigues I. Co-investigators: AbdelMaguid Sabsabi LA, Barnett TA, Brousseau-Foley M, Bériault K, Courville-LeBouyonnec A, Dorimain VM, Gauthier A, Glaser E, Gonzalez Reyes A, Groulx A, Isabel M, Lambert LG, Larney S, Lussier MT, Lussier S, MacLaren L, Motard A, **Petrakos NZ**, Pétrin-Desrosiers C, Schuster T, Semret M, Sergerie V, Talbot A, Todd KJ Collaborators: Bosoi MC, Boucher MC, Charbonneau V, Cohen M, Delage B, Dubé A, Dufresne-Tassé C, Langevin S, Layani G, Mayamona YKK, Morin C, Picotte ME, Simard E, Trépanier E, Valcourt E.	2026 – 2027
CIHR Project Grant Competition – Spring 2026 (submitted, $1,075,000) Evaluating the implementation and real-world effectiveness of statin-based cardiovascular primary prevention in routine HIV care Principal Investigator: de Pokomandy A (nominated) Co-investigators: Durand M, Fontaine G, Fortin E, Mansour S, Marbaniang I, Martel-Laferrière V, Monteith K, **Petrakos NZ**, Sheehan N, Thomas R, Vachon ML Collaborators: Bakombo D, Baril JG, Depatureaux-Gérémy A, Falutz J, Levy L, Pilarski R	2026 – 2031

CIHR Project Grant Competition – Spring 2026 (submitted, $1,005,000) 2026 – 2031
Understanding and Improving Acute Respiratory Infection Care Trajectories and Antibiotic Use in Primary Care using the COhort in PRImary Care (COPRI)
Principal Investigator: de Pokomandy A (nominated)
Co-Prinicipal Investigators: Gaboury I, Paquette JS, Rodrigues I
Co-investigators: Barnett TA, Bériault K, Brousseau-Foley M, Courville-LeBouyonnec A, Dorimain VM, Gauthier A, Glaser E, Gonzalez Reyes A, Groulx A, Isabel M, Lambert LG, Layani G, Lussier MT, Lussier S, MacLaren L, Motard A, **Petrakos NZ**, Pétrin-Desrosiers C, Racine-Hemmings F, Schuster T, Semret M, Sergerie V, Talbot A, Todd KJ, Wanneh E
Collaborators: Boucher MC, Charbonneau V, Bosoi MC, Cohen M, Delage B, Dubé A, Dufresne-Tassé C, Mayamona YKK, Langevin S, Morin C, Picotte M, Simard E, Trépanier E, Valcourt E
